

AGENEWTECH

H1502RQ

Smart Module Specification_V1.3



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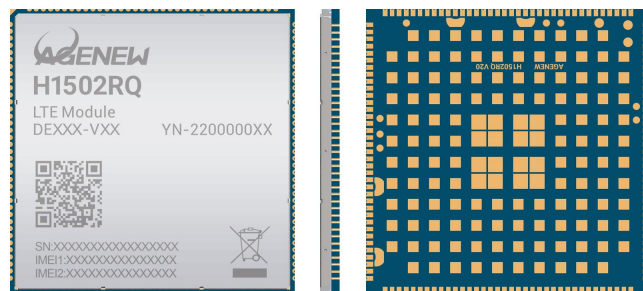
About the Document

History

Version	Data	Author	Description
-	2024-05-30	Wu Danyi	Initial
1.0	2024-06-15	Wu Danyi	Controlled Copy
1.1	2025-05-28	Wu Danyi	Modify frequency band information
1.2	2025-05-29	Wu Danyi	Change the file name from H1502RP to H1502RQ; Other format optimizations
1.3	2026-03-23	Chen Yun	Change company name

H1502RQ

Multi-network Smart LTE module Supports Wi-Fi & BT



H1502RQ is a multi-network Smart LTE Cat 4 module based on MediaTek's Octa-core (Dual-core ARM Cortex-A75 + Six-core ARM Cortex-A55) processor launched by AGENEW, equipped with Android 14 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

H1502RQ supports full coverage of all network systems, WI-FI 802.11a/b/g/n/ac, and BT5.0 near-field wireless communication. In addition, the H1502RQ also has audio and video input and output interfaces and rich GPIO interfaces. It also supports a variety of audio and video codecs. The home screen supports FHD+2520*1080 at most. At the same time, H1502RQ supports GPS/GLONASS/BDS/Galileo positioning systems, making its positioning function fast and accurate.

H1502RQ is a patch module with a total of 274 pins, including 146 LCC pins and 128 LGA pins. The size is only 40.5mm × 40.5mm × 2.8mm, which can be embedded in various M2M product applications through pads. H1502RQ integrates rich interfaces (such as LCM, camera, touch screen, microphone, speaker, UART, USB, I2C, SPI, SDIO, and ADC interfaces), greatly expanding its application in the M2M field, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle equipment, and high-end information Collection equipment, smart robots, smart homes, smart hardware, industrial smart handheld devices, drones, high-end police or law enforcement equipment, smart intercom devices, smart wearables, vending machines, logistics cabinets and other equipment are used in different industries.



Main Advantages

- ✓ Comprehensive coverage of multi-network standard
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT5.0 short-range wireless communication
- ✓ Support GPS/Glonass/BDS/Galileo
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to FHD+ 2520×1080
- ✓ Support 1080P@30fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



Octa-core ARM A75+A55
64-bit processor



IEEE 802.11
a/b/g/n/ac



Support Android 14



GPS+Glonass+BDS+Galileo



Powerful multimedia



BT5.0

LTE Cat4	H1502RQ	
Region/Operator	Eurasia/South America	
Operating System	Android 14	
Memory	32GB eMMC + 3GB LPDDR4X/32GB eMMC + 4GB LPDDR4X/64GB eMMC + 4GB LPDDR4X	
Module size (mm)	40.5 × 40.5 × 2.8	
Packaging	LCC + LGA	
Temperature Range		
Operation Temperature	Normal operating temperature: -20 °C ~ + 80°C	
	Storage temperature range: -40 °C ~ + 90 °C	
Frequency Band Information	P3	P4
LTE-FDD	B1/B2/B3/B4/B5/B7/B8/B28A/B28B	B1/B3/B5/B7/B8/B20/B28A/B28B
LTE-TDD	B40	B38/B40/B41(194M)
WCDMA	B1/B2/B5/B8	B1/B5/B8
GSM/EDGE	B2/B5	B3/B5
WLAN	2.4/5GHz, 802.11 a/b/g/n/ac	
Bluetooth	5.0	
GNSS	GPS + GLONASS + BDS + Galileo	
Authentication		
Operator Certification	Undetermined	
Compulsory Certification	Undetermined	
Other Certifications	Undetermined	
Data Rate		
LTE-FDD (Mbps)	300Mbps (DL) / 75Mbps (UL)	
LTE-TDD (Mbps)	260Mbps (DL) / 90Mbps (UL)	
DC-HSPA+ (Mbps)	42Mbps (DL) / 11.2Mbps (UL)	
WCDMA (Kbps)	384Kbps (DL) / 384Kbps (UL)	
EDGE (Kbps)	296Kbps (DL) / 236.8Kbps (UL)	
GPRS (Kbps)	107Kbps (DL) / 85.6Kbps (UL)	
Outstanding characteristics		
Five-Mode	LTE: Release 12 UMTS: Release 9	
Bandwidth (MHz)	1.4/3/5/10/15/20	
Rx-Diversity	●	
DL supports MIMO 2 × 2	●	
SIM Card Detection	●	
Dual SIM Dual Standby	●	
USB Firmware Upgrade	●	
OTA Firmware Upgrade	●	

Remarks: 1. *: In progress

2. ●: Supports

3. Extended operating temperature: If the module works in this temperature range, its RF performance might be worse but it can still work properly.

LTE Cat4	H1502RQ
Electrical specification	
Operating Voltage	3.5~4.4 V, typical 3.8 V
Consumption (typical)	Undetermined
Interface	
LCD	4-lane MIPI_DSI, FHD+ 2520×1080@60fps
Camera	Support 3 sets of camera, up to 25MP
TP	Capacitive touch screen
Audio	×3 Analog inputs: MIC1, MIC2, HEADMIC (×1 noise cancelling microphone) ×3 Analog outputs: differential linear (requires external audio amplifier), differential receiver and stereo earphone
USB	USB 2.0, Supports OTG, Charge
(U)SIM	2 sets, Supports 1.8/3V (U) SIM card, supports (U) SIM card hot plug detection, dual card dual standby
UART	2 UARTs for debugging and applications
SD Card	1 set
ADC	2 sets, 1 set general ADC interface, 1 set for battery ID
I2C	7 I2Cs to control peripheral devices
SPI	4 SPI interface
PWRKEY	Internal pull up to VBAT
GPIO	18 GPIO reserved
ANT	Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna